Italics

WHEN NOT TO USE ITALICS

1) Do not use italics for emphasis or for defining words within the text, references, or elsewhere.

2) Do not use italics for well-known Latin words (or their abbreviations), nor for other foreign words.

in vivo, in situ, e.g., i.e., bremsstrahlung

(However, we do italicize *et al*.)

3) Do not use italics for semiconductor types or atomic configurations:

n-type semiconductor, p-type semiconductor
sp^2-bonded carbon
s-, p-, and d-wave (quantum physics/semiconductor/superconductivity papers)

4) Do not italicize commas separating items in a list of italic terms.

Escherichia coli, *Bacteroides ovatus*, and *Butyricimonas virosa* isolates

5) Do not italicize parentheses or brackets surrounding italic terms. (The only exceptions are for terms such as genotypes that may include parentheses.)

USE ITALICS IN THE FOLLOWING CASES:

1) **Chemical Terms**

- Positional, structural, and configurational prefixes (for example, *cis*, *sec*, *tert*, *trans*, *meso*, *sym*, *m*, *n*, *o*, *p*) when used with a chemical name or formula. If terms such as these are used without a chemical name, do not italicize them.

p-benzenediacetic acid
o-dibromobenzene
n-butyl iodide
sec-butyl alcohol
but: a cis or trans configuration

Follow author style in abbreviating such terms once introduced.

m-chloroperbenzoic acid (*m*CPBA)

With certain classes of compounds that are not chemicals (technically speaking), editors should follow the author's style.

> The *cis*-2,3-diols react faster than. . . ["diols" is not a chemical name (it is a class of chemicals), although it is used here as if it is]

- Stereoisomer descriptors (*cis*, *trans*, *meso*, *exo*, *rel*, *endo*). (See *ACS Style Guide* for further examples)

> *d*-camphor
> *meso*-tartaric acid

Note: The prefixes cis and trans, which are italicized in chemical terms, are not italicized when used with genetic terms.

> trans-acting

- Letters in square brackets in names of polycyclic aromatic compounds.

> 1*H*-benz[*de*]naphthacene

- In polymer terms, italicize *alt, b*, *co*, *g*, *r*, and *m* when used with a chemical name.

> poly(styrene-*g*-acrylonitrile)

- The letters in symmetry and space groups, but not the numbers.

> $Fd3m$
> C_{2v}

- Symbols of chemical elements that show attachment to an atom. Italicize these symbols when they are part of a chemical name.

> *N*-ethylaniline
> 3*H*-fluorine
> *N*-methylcystine
> *N*,*N*´-dimethylurea
>
> But do not italicize if there is no specific attachment to an atom:
>
> N-glycosylated form
> N-linked, O-linked, S-linked

- Enzymes

> *N*-ribosyl-purine ribohydrolase
> *O*-methyltransferase
> *S*-adenosylmethionine

However, acronyms for such terms are all roman.

glutathione *S*-transferase (GST)

(For a detailed explanation and listing of examples, see the *ACS Style Guide*)

2) **Foreign Words**

- Unfamiliar foreign terms particular to the subject of a paper (for example, given with a translation) should be italicized.

…the interference of the positive- and negative-energy components of the wave packet (*Zitterbewegung*)…

- In a quotation, *sic* is in brackets and italicized.

"The satelite [*sic*] was launched. . . "

(See *Chicago* for examples and advice on determining the familiarity of certain words. *Words into Type* presents a list of foreign words, which is helpful for checking spelling, but many terms are shown in italics and do not follow *Science* style.)

3) **Genetics**

Italicize	Set roman
alleles	antigens
names of specific genes	names of specific proteins
	restriction endonucleases
	Hind II, Eco RI

RNA molecules are often described in terms of Svedberg units (sedimentation coefficients); the *S* is italic, but not the number of units:

the universal 16*S* rRNA primer sequence

Plants:
Arabidopsis thaliana:
-Genes: symbols consist of three (occasionally two) italicized uppercase letters, e.g., *EMB*, *DET*. Different loci with the same three-letter symbol are distinguished by different numbers, e.g., *EMB1*, *EMB2*, etc.
-Proteins: designation in capital letters and not italicized, e.g., DET1.

Yeast:
S. cerevisiae:

-Genes: three italic letters followed by an italicized arabic numeral, e.g., *ARG2*, *arg2* *ade5*, *cdc28*. On the genetic map, the locus is named after the mutant phenotype, and because most mutations studied are recessive, almost all locus designations are lowercase, e.g., *arg2*.
-Proteins: same characters as gene symbol in roman type, with initial capital letter (Arg); superscript plus or minus indicates the independence of, or requirement for, a substance, respectively, e.g., Arg^+ [does not require arginine]

C. elegans
Genes: 3 lowercase italic letters, a hyphen, and an arabic no.,e.g., *dpy-5*, *let-37*
Proteins: the relevant gene name in nonitalic capital letters, e.g., the protein encoded by *unc-13* is UNC-13

Drosophila:
genes:
-symbols vary depending on whether allele is recessive to wild type (begins with lowercase letter, e.g., *awd* for *abnormal wing disc*) or dominant to wild type (begins with uppercase letter, e.g., *R* for *Roughened*)
-Genes named after a protein begin with an uppercase letter, eg. *Adh* (for alcohol dehydrogenase)
-different loci with similar function: gene symbol plus italic arabic numerals and capital letters; hyphens not used (e.g. *Act5C*, *Act42A*)
-Proteins: same as gene symbol but in nonitalic capital letters, e.g. HH (protein encoded by *hh* gene)

Mouse
Genes: 2, 3 or 4 italic letters and numbers, e.g. *Re*, *G6pd*, *B2m*; the first letter is uppercase.
Proteins: same symbol as the gene, all-uppercase, no italics; e.g., the product of the *Myc* gene is MYC.

Human:
Genes: uppercase italic letters or combination of uppercase letters + arabic numerals (*ADH*, *ACO1*); Greek letters, roman numerals, superscripts, subscripts, and punctuation are not acceptable
Proteins: same characters used for the loci encoding them, in nonitalic type

(See *CSE Style Manual* for a detailed discussion of genetic nomenclature)

Here are online resources for more specific genetic and protein guidelines.
The Arabidopsis Information Resource Nomenclature Guidelines
http://www.arabidopsis.org/portals/nomenclature/guidelines.jsp
UniProtKB/Swiss-Prot nomenclature http://www.expasy.ch/cgi-bin/lists?nomlist.txt
Human Gene Nomenclature Committee http://www.genenames.org/guidelines.html
Human Genome Variation Society http://www.hgvs.org/mutnomen/recs.html
Genetic Nomenclature for *Drosophila melanogaster*
http://flybase.org/wiki/FlyBase:Nomenclature

International Union of Pure Applied Chemistry
http://old.iupac.org/publications/epub/index.html#nt
RCSB Protein Data Bank http://www.rcsb.org/pdb/home/home.do
Mouse Genome Informatics http://www.informatics.jax.org/mgihome/nomen/gene.shtml
Candida Genome Database http://www.candidagenome.org/Nomenclature.shtml
Saccharomyces Genome Database http://www.yeastgenome.org/
Genetic (and organism) nomenclature for bacteria http://jb.asm.org/site/misc/journal-ita_nom.xhtml

4) **Genus and Species**

Italicize genus, species, and subspecies names of plants, animals, and microorganisms.

Acer saccharum
Drosophila mauritiana
Escherichia coli

However, do not italicize:

- Genus when it is used in its adjectival form.

Ex: *Streptococcus*, but streptococcal

- Name of person given after a genus and species

Felis leo Scop. (abbreviated name)

- Abbreviations such as sp. and var.

- Larger divisions, such as family and phylum (Arthropoda, Nematoda)
But names of all bacterial taxa (kingdoms, phyla, classes, orders, families, genera, species, and subspecies) should be italicized.

5) **Math and Statistics**

Italicize variables, constants, and unknown quantities in the text and in equations (see *ACS Style Guide*).

V - voltage
I - current
T - temperature
t - time

Italicize quantum physics terms:

a spin $S = ½$ iron(V)-oxo complex

Do not italicize: - Vectors (set in boldface roman)
- Abbreviations for operators (cos, exp, lim, log, sin)
- Greek letters (always set in roman type)
- but: log *x*, exp *y*

(See *ACS*)

Other examples:

n - number of items

*n*th power, *i*th mode

F - degrees of freedom

$F_{7,105} = 2.95$

f(*x*) - function of *x*

P - probability

$P > 0.01$

t test, *U* test

$dy/dx = nx^{n-1}$ - differential coefficient

dx - total differential of *x*

Superscripts and subscripts (of terms in italics, or of other terms) should be set roman, unless introduced as an italic variable elsewhere.

(See list of symbols in *ACS*)

6) **Units of Measure and Other Abbreviations**

A list of some common abbreviations in italics follows:

g – acceleration equivalent to one gravity, as when a centrifuge is used
20,000*g* (not 20,000X*g*)

e - charge on an electron

F_{obs}, F_{calc} - observed and calculated structure factors in crystallography

k - in general, a rate constant

K - in general, an equilibrium constant, usually with a roman subscript

K_{m} - Michaelis constant

A - absorbance, with numeral subscript indicating wavelength (A_{260})

M - magnitude (earthquake)

$L_{\odot}$ - luminosity of the Sun

$M_{\odot}$ - mass of the Sun

S wave and *P* wave (earthquake papers)

7) **References**

Italicize:

- Titles of books, periodicals, and newspapers

- *et al*. (after an author's name in the reference list and in the text)

- Reference citations in text

The extent of acylation was assayed (*7*).

8) **Other Uses**

Italicize titles or names of

- Publications

- Ships

- Plays, movies, poems, operas, and long musical compositions

- Works of art

Note: When a proper noun is italicized, the possessive ending is roman.

the *Saturday Review*'s editor

Do not italicize, per *Science* style:

- Spacecraft (Pioneer 1, Voyager 1)

- TV and radio programs

When naming legal cases, italicize names of plaintiff and defendant, but not the "v."

> *Smith* v. *Maryland*

9) Typography

If a word or phrase would normally be set italic but appears as part of a heading or other part of a manuscript that is already in italics, set the word or phrase in roman type to distinguish it, as in the book title

> *The History of* C. difficile *in Scotland*
> Arabidopsis*: A Laboratory Manual*